



UNIVERSITÉ  
**PARIS  
SCIENCES  
& LETTRES**

# Statistical Learning II

Lecture 1 - Introduction & preliminaries

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**Bruno Loureiro**  
@ CSD, DI-ENS & CNRS

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# Course Organisation

- 12 classes of 3h, divided into:
  - 1h30 lectures
  - 1h30 exercises & lab (Python)  
(with **Alejandro Saldarriaga-Fuertes**)



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- Contact us: via **Microsoft Teams** or **e-mail**:

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- Evaluation: 2 exams (**midterm** and **final**)

# Menu for the semester

**Goal**: Develop a *mathematical* understanding of *classical* and *modern* machine learning models

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- Classical methods:
  - Ridge regression
  - LASSO
  - Generalised linear models
  - Kernel methods
  - Principal component analysis (PCA)

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- Classical methods:
  - Ridge regression
  - LASSO
  - Generalised linear models
  - Kernel methods
  - Principal component analysis (PCA)
- Modern methods:
  - Neural networks
  - Diffusion models
  - Your suggestions?

# Introduction & Motivation

*Or: why should I care about Statistical Learning?*

---

# A brief history of ML

W. McCulloch W. Pitts



McCulloch-Pitts  
neuron  
1943

F. Rosenblatt



Rosenblatt's  
perceptron /  
MLP  
1957

S. Linnainmaa



Backprop  
1970

K. Fukushima



Neocognitron  
1980

Y. LeCun



CNNs  
1989



2016  
AlphaGo



ChatGPT  
2022

1940

1st digital  
computer

1947

Invention of the  
transistor

1969

ARPANET  
(Internet)

1982

Hopfield  
Model

1997

IBM Deep  
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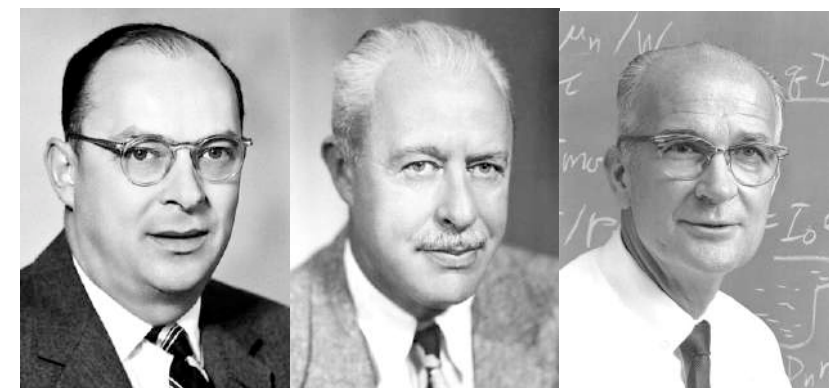
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J. Bardeen W.H. Brattain W. Shockley



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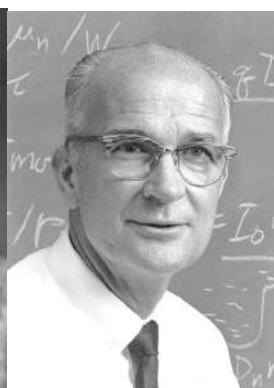
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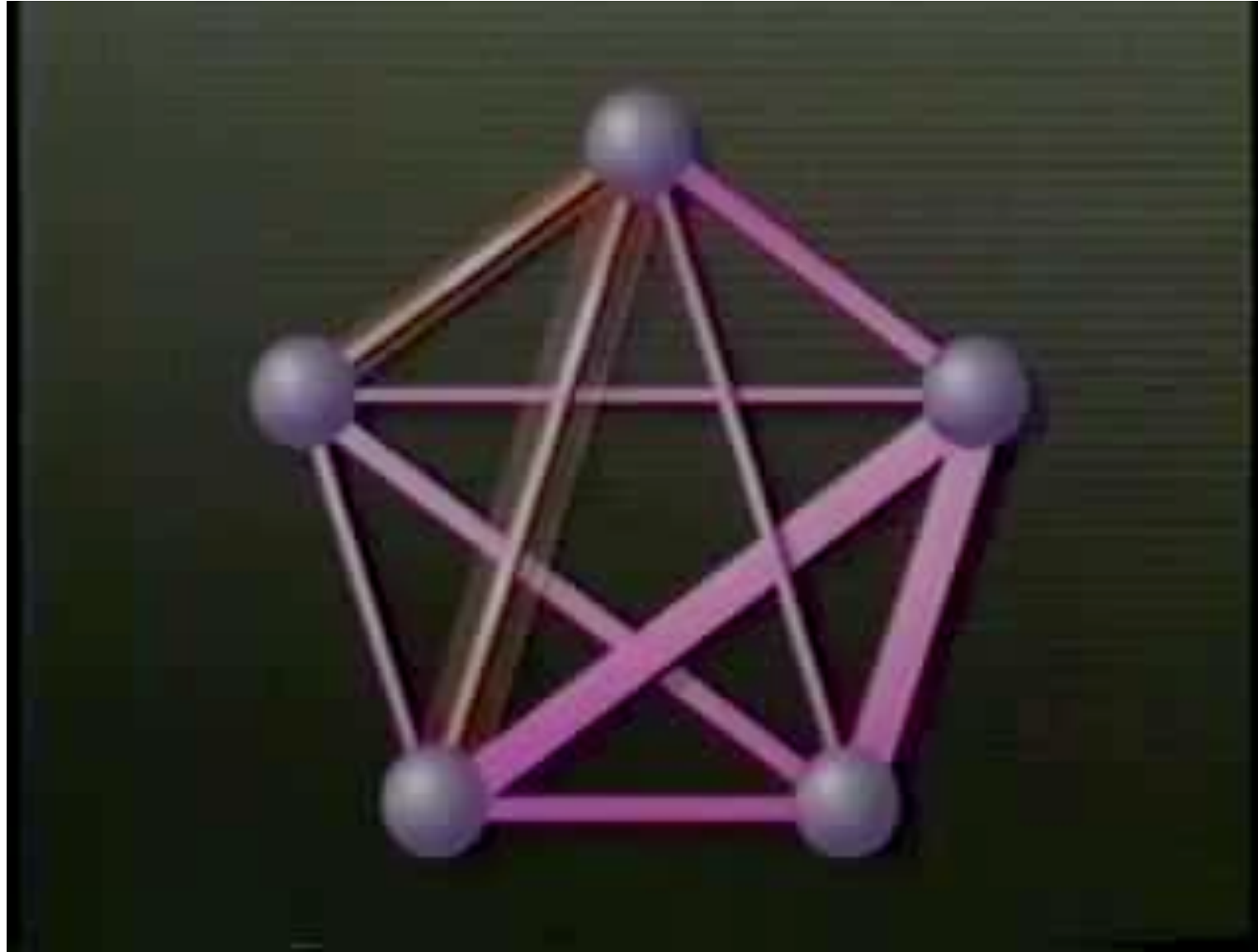
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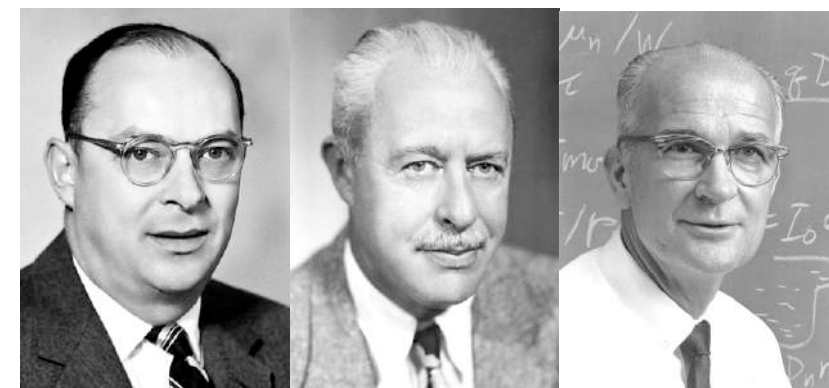
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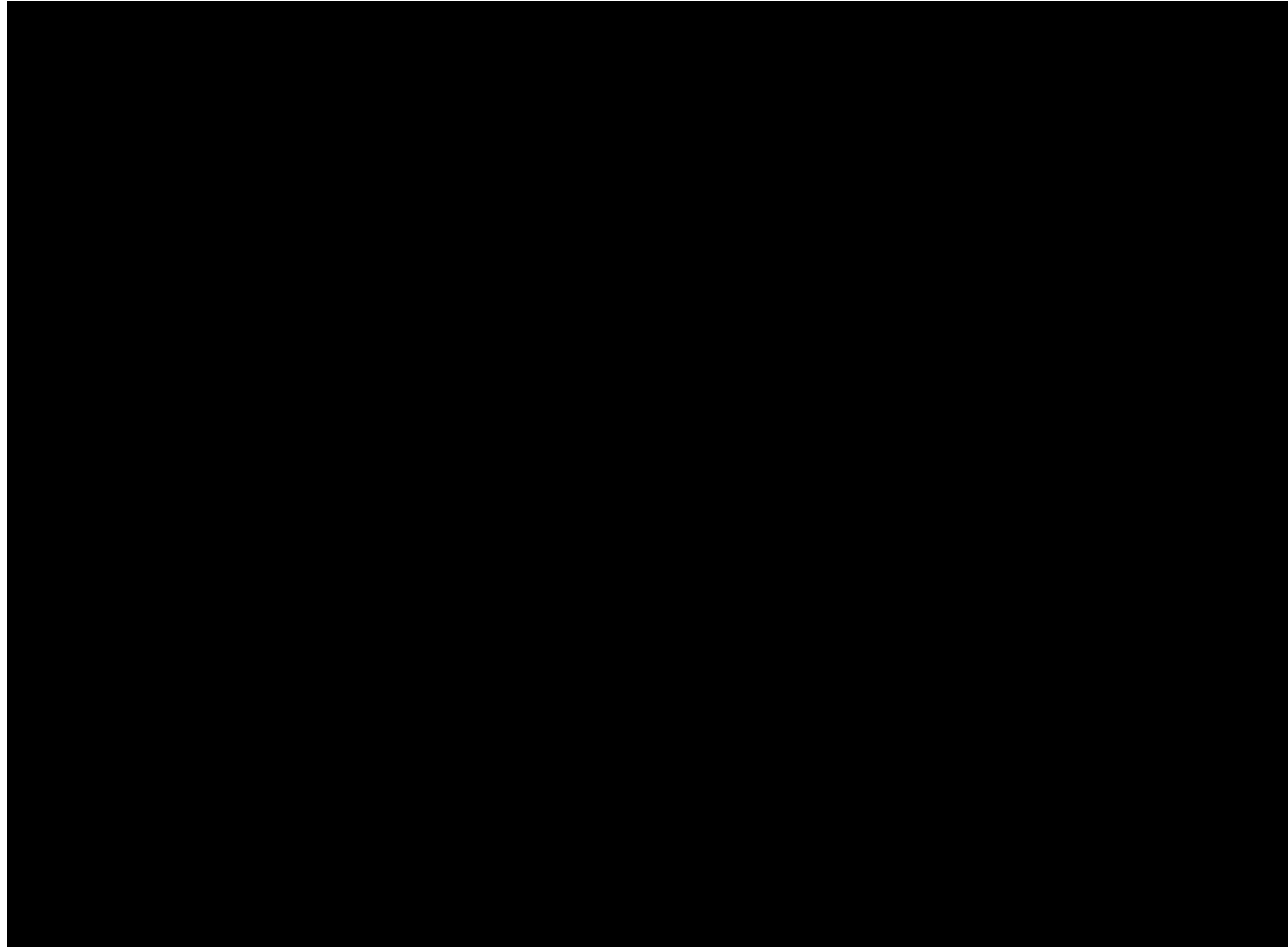
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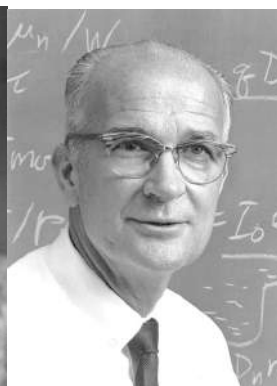
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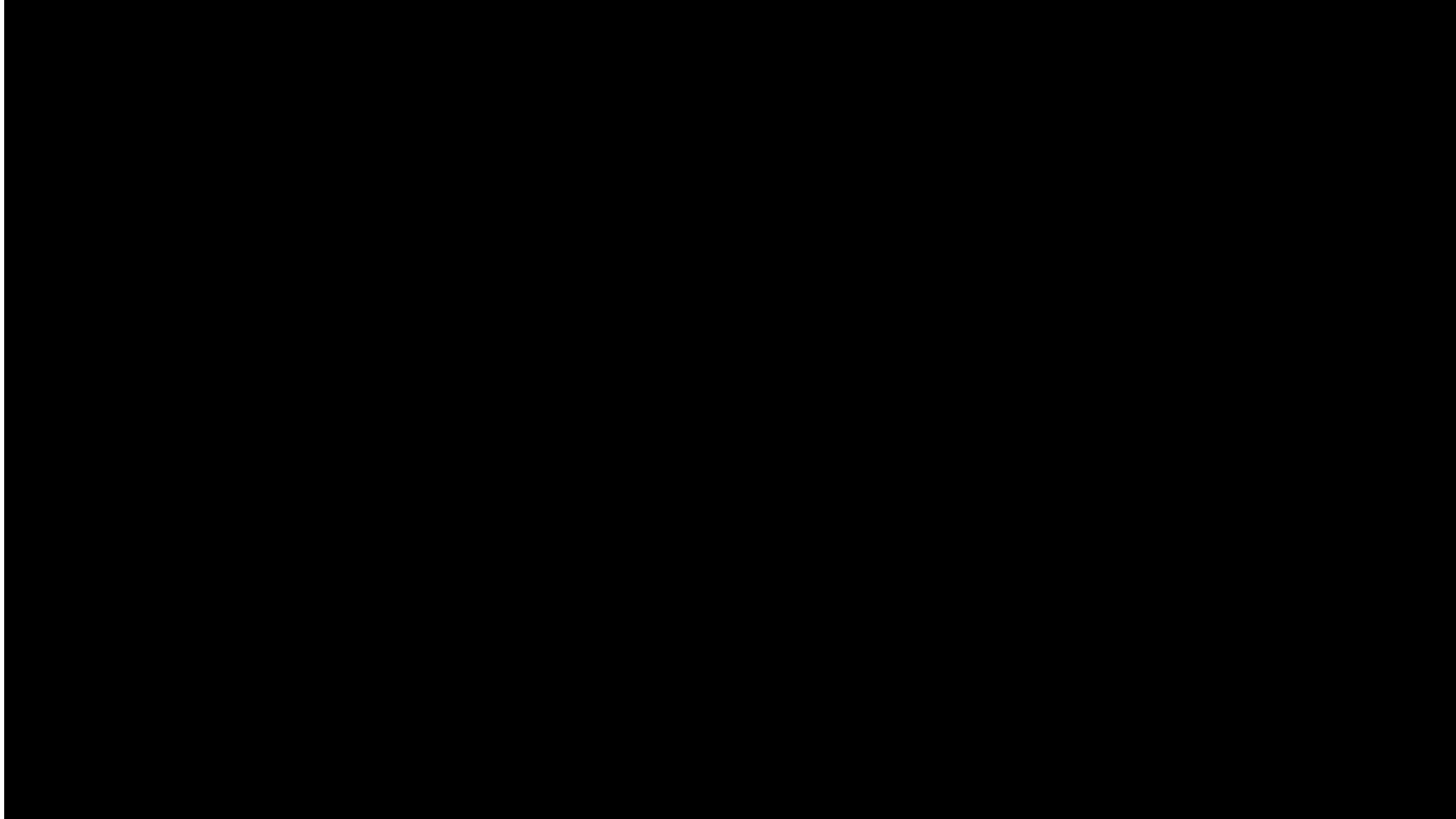
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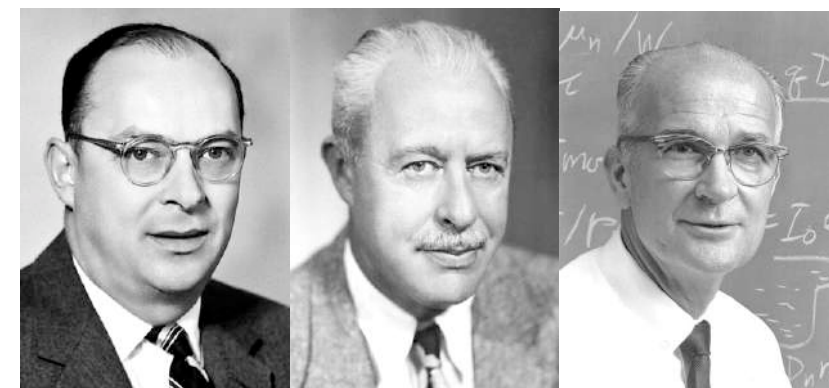
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# What about the maths?

Yet, on the mathematical side...

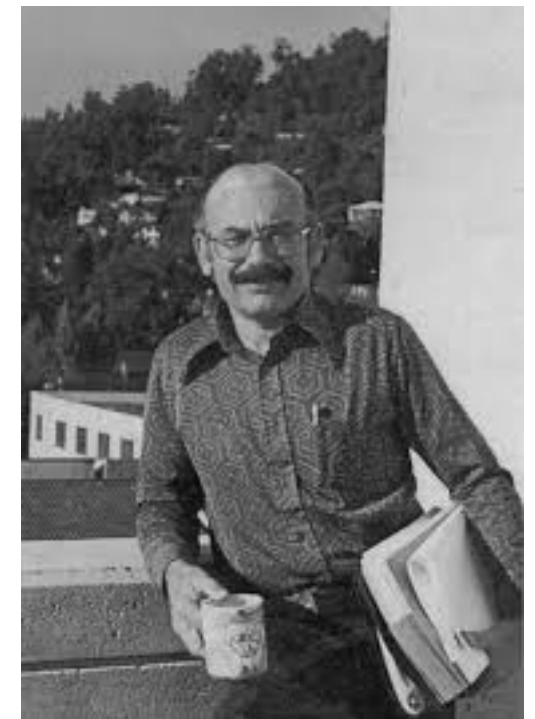
## **Leo Breiman**

Statistics Department, University of California, Berkeley, CA 94305;  
e-mail: [leo@stat.berkeley.edu](mailto:leo@stat.berkeley.edu)

## Reflections After Refereeing Papers for NIPS

For instance, there are many important questions regarding neural networks which are largely unanswered. There seem to be conflicting stories regarding the following issues:

- Why don't heavily parameterized neural networks overfit the data?
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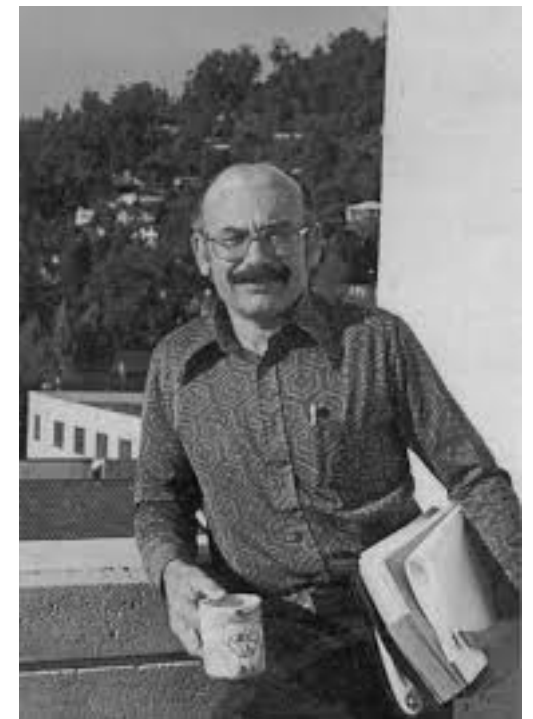
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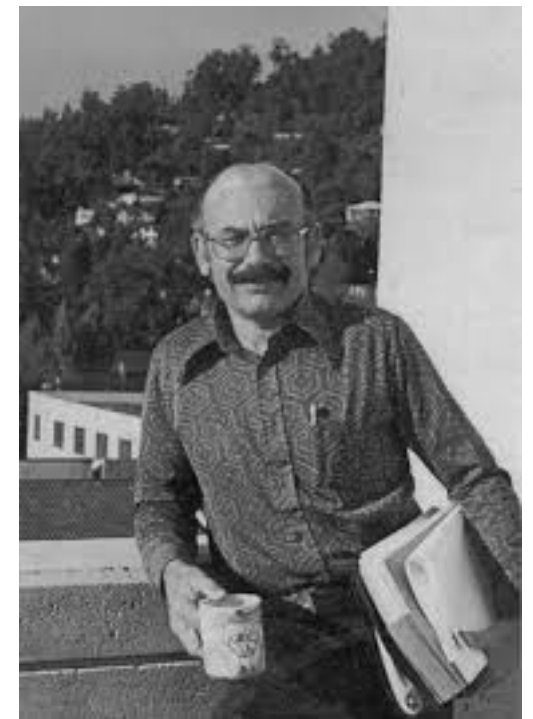
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But why should I care?

# Some good reasons to care

- Reliability and Liability

If a model does something unexpected, **who is responsible?**

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Data centres are responsible for **4% of the energy consumption in the US.**

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Can we **design models** and **algorithms** that learn more **efficiently** from data?

- Scientific curiosity

- And in the worst case, understanding the maths will make you a **better engineer / data scientist.**

# Our expectations

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**My expectations:** By the end of the term, I expect you to:

- Get acquainted with the most popular machine learning algorithms
- Appreciate (some) of the mathematics behind the methods.
- Be able to implement these methods from scratch.

# Recap of Linear Algebra

The bread of statistical learning

---

# The Euclidean space

The Euclidean space  $\mathbb{R}^d$  is the vector space of  $d$ -tuples:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix} \in \mathbb{R}^d \quad (\mathbb{R}^{d \times 1})$$

“column vector”

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Recall,  $\mathbb{R}^d$  is a vector space of dimension  $d$  with basis:

$$\mathbf{e}_i = \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix}$$

Position  $i$



# The Euclidean space

The Euclidean space is endowed with an **inner** (or **scalar**) product

$$\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^d \qquad \langle \boldsymbol{u}, \boldsymbol{v} \rangle = \sum_{i=1}^d u_i v_i$$

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$$\mathbf{u}, \mathbf{v} \in \mathbb{R}^d \qquad \langle \mathbf{u}, \mathbf{v} \rangle = \sum_{i=1}^d u_i v_i$$

Which induces a natural notion of **distance** and **size**:

$$\|\mathbf{u}\|_2 = \sqrt{\langle \mathbf{u}, \mathbf{u} \rangle} = \sqrt{\sum_{i=1}^d u_i^2} \qquad d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|_2$$

“Euclidean or  $\ell_2$  norm”                      “Euclidean distance”

We say two vectors  $\mathbf{u}, \mathbf{v} \in \mathbb{R}^d$  are **orthogonal** if  $\langle \mathbf{u}, \mathbf{v} \rangle = 0$

# Euclidean geometry

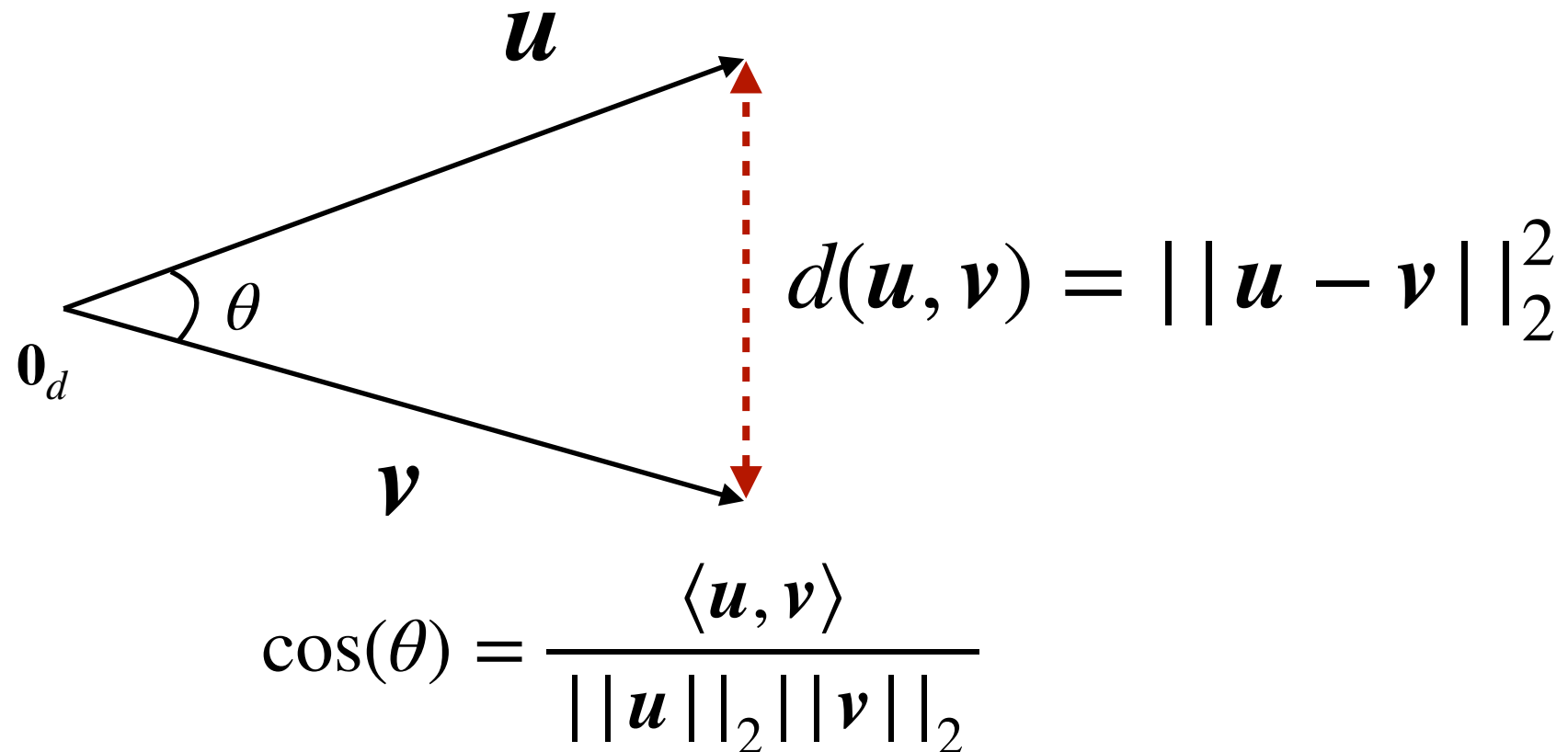
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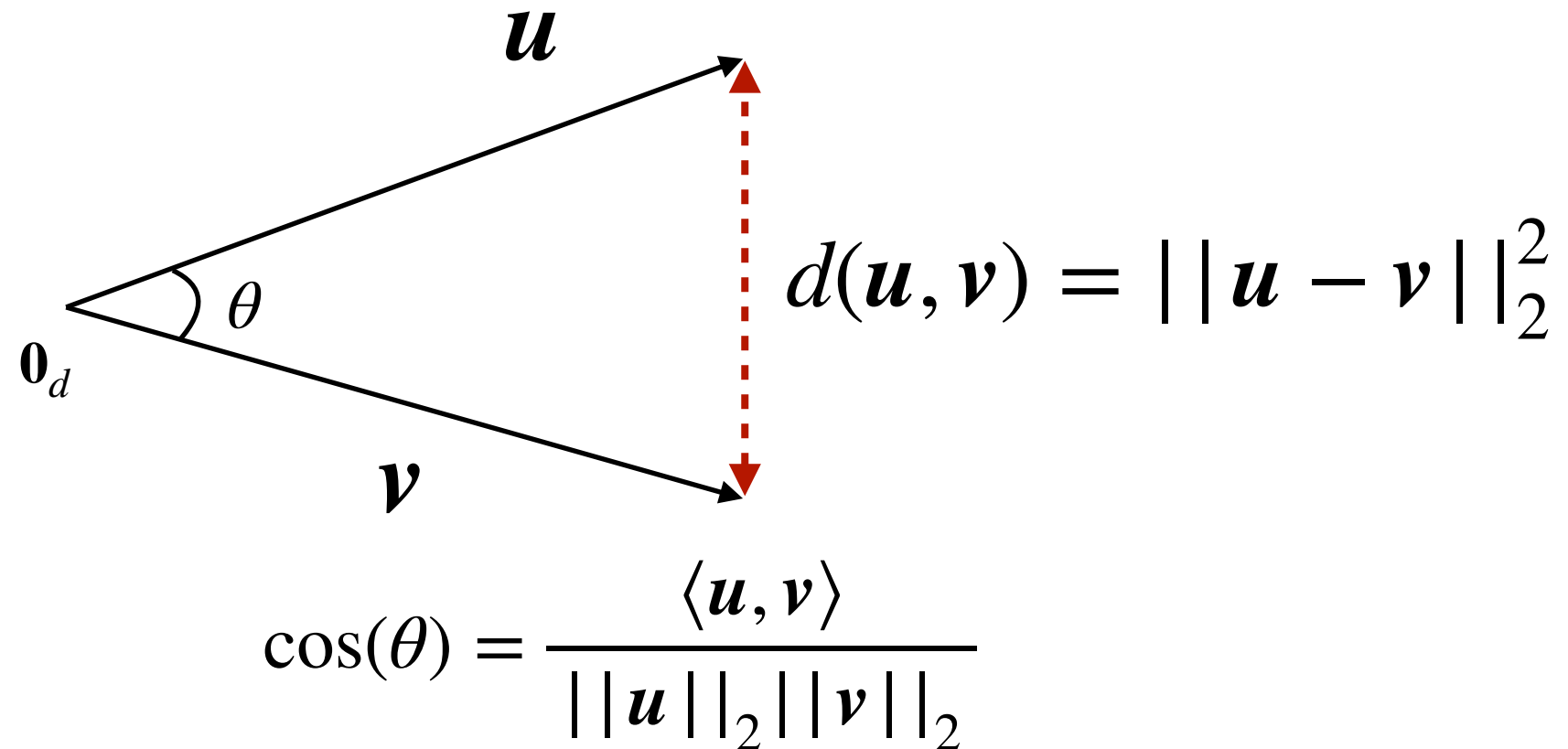
“Euclidean distance”

They correspond to our intuitive notion of geometry in the plane



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In particular, we say two vectors  $\mathbf{u}, \mathbf{v} \in \mathbb{R}^d$  are **orthogonal** if

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# Other norms

One can define other notions of size in  $\mathbb{R}^d$

$$||\mathbf{u}||_p = \left( \sum_{i=1}^d u_i^p \right)^{1/p} \quad p \geq 1$$

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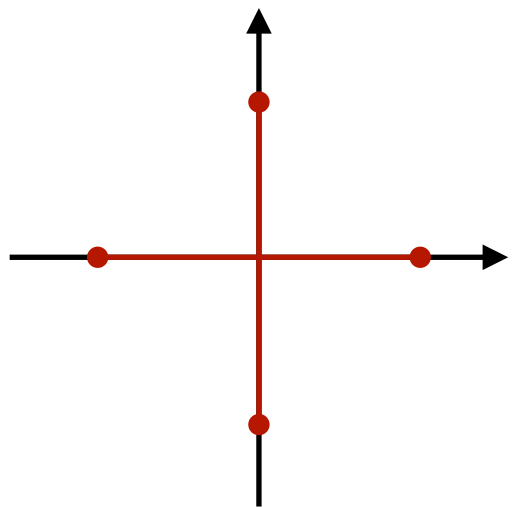
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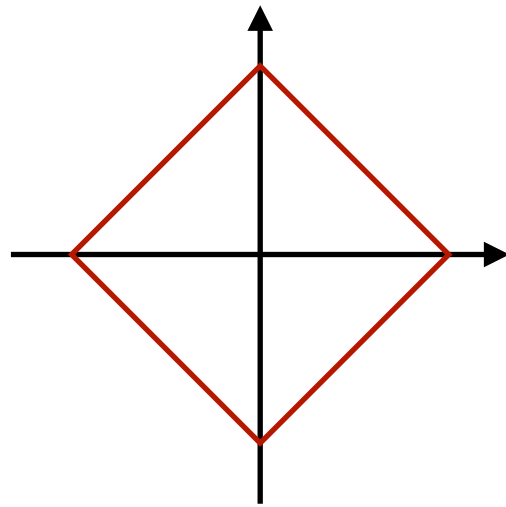
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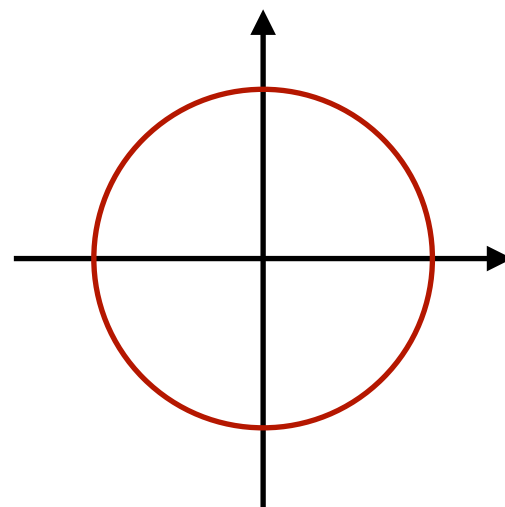
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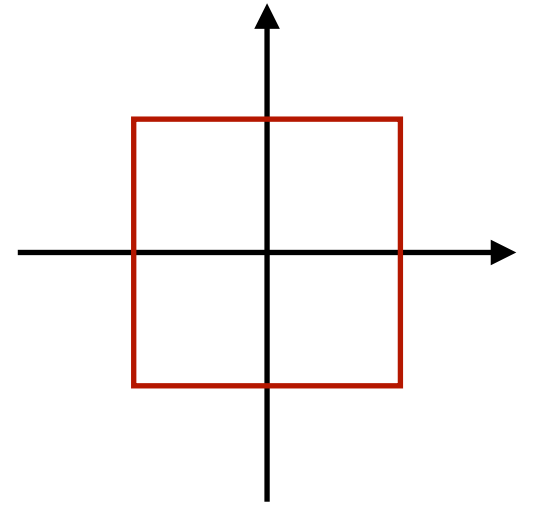
$\ell_0$



$\ell_1$



$\ell_2$



$\ell_\infty$



Not a norm

# Matrices

---

A real-valued matrix  $\mathbf{A} \in \mathbb{R}^{n \times d}$  is a table of real numbers.

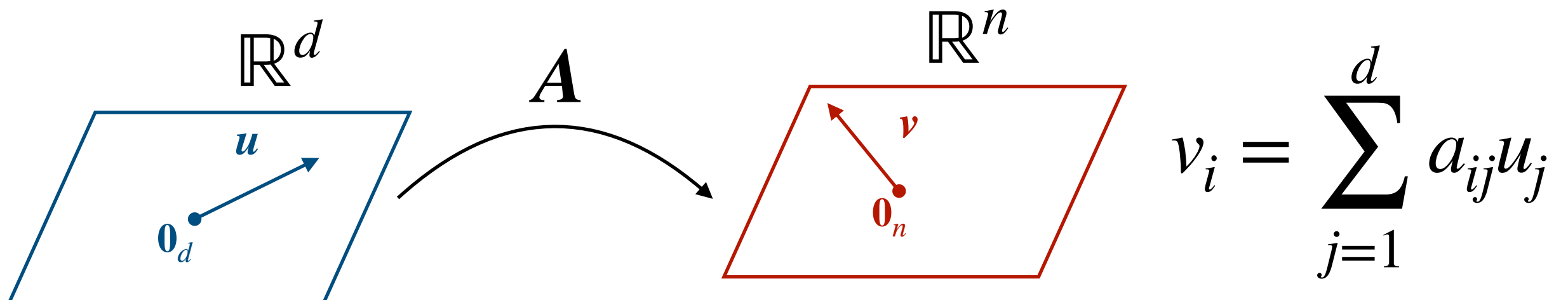
$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1d} \\ a_{21} & a_{22} & \cdots & a_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nd} \end{bmatrix} \in \mathbb{R}^{n \times d}$$

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It is most often used to describe the coordinates of linear transformations  $\mathbf{A} : \mathbb{R}^d \rightarrow \mathbb{R}^n$  with respect to a basis.



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$\mathbf{A}_1 \quad \mathbf{A}_2 \quad \cdots \quad \mathbf{A}_d$

- The columns of  $\mathbf{A} \in \mathbb{R}^{n \times d}$  are vectors  $\mathbf{A}_i \in \mathbb{R}^n$  with  $(\mathbf{A}_i)_j = a_{ij}$
- “Column space”  $\text{col}(\mathbf{A}) = \text{span}(\mathbf{A}_1, \cdots, \mathbf{A}_d) \subset \mathbb{R}^n$

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- The rows of  $\mathbf{A} \in \mathbb{R}^{n \times d}$  are vectors  $\mathbf{a}_j \in \mathbb{R}^d$  with  $(\mathbf{a}_j)_i = a_{ij}$

“Row space”  $\text{row}(\mathbf{A}) = \text{span}(\mathbf{a}_1, \dots, \mathbf{a}_n) \subset \mathbb{R}^d$

# Flattening matrices

The space of matrices  $\mathbf{A} \in \mathbb{R}^{n \times d}$  is itself a vector space of dimension  $nd$ . Therefore we can identify:

$$\mathbb{R}^{n \times d} \simeq \mathbb{R}^{nd}$$

By **flattening** the matrices into vectors.

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nd} \end{bmatrix} \mapsto \begin{bmatrix} a_{11} \\ a_{12} \\ \vdots \\ a_{1n} \\ a_{21} \\ \vdots \end{bmatrix} \in \mathbb{R}^{nd}$$

# Rank of a matrix

- The **rank** of a matrix  $A \in \mathbb{R}^{n \times d}$  is the dimension of column space

$$\text{rank}(A) = \dim(\text{col}(A))$$

This is equivalent to the **number of independent columns**.

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**Proposition**

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## Proposition

$$\text{rank}(A) = \dim(\text{col}(A)) = \dim(\text{row}(A))$$

- A matrix  $A \in \mathbb{R}^{n \times d}$  is said to be **full-rank** if

$$\text{rank}(A) = \min(n, d)$$

# Another point of view

- Alternatively, we can see the **column space**  $\text{col}(\mathbf{A}) \subset \mathbb{R}^n$  as The **image** of the associated linear map.

$$\text{im}(\mathbf{A}) = \text{col}(\mathbf{A}) = \{ \mathbf{v} \in \mathbb{R}^n : \mathbf{A}\mathbf{u} = \mathbf{v} \text{ for some } \mathbf{u} \in \mathbb{R}^d \}$$

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- The **null-space** or **kernel** of a matrix  $\mathbf{A} \in \mathbb{R}^{n \times d}$  is defined as:

$$\text{ker}(\mathbf{A}) = \{ \mathbf{u} \in \mathbb{R}^d : \mathbf{A}\mathbf{u} = \mathbf{0} \}$$

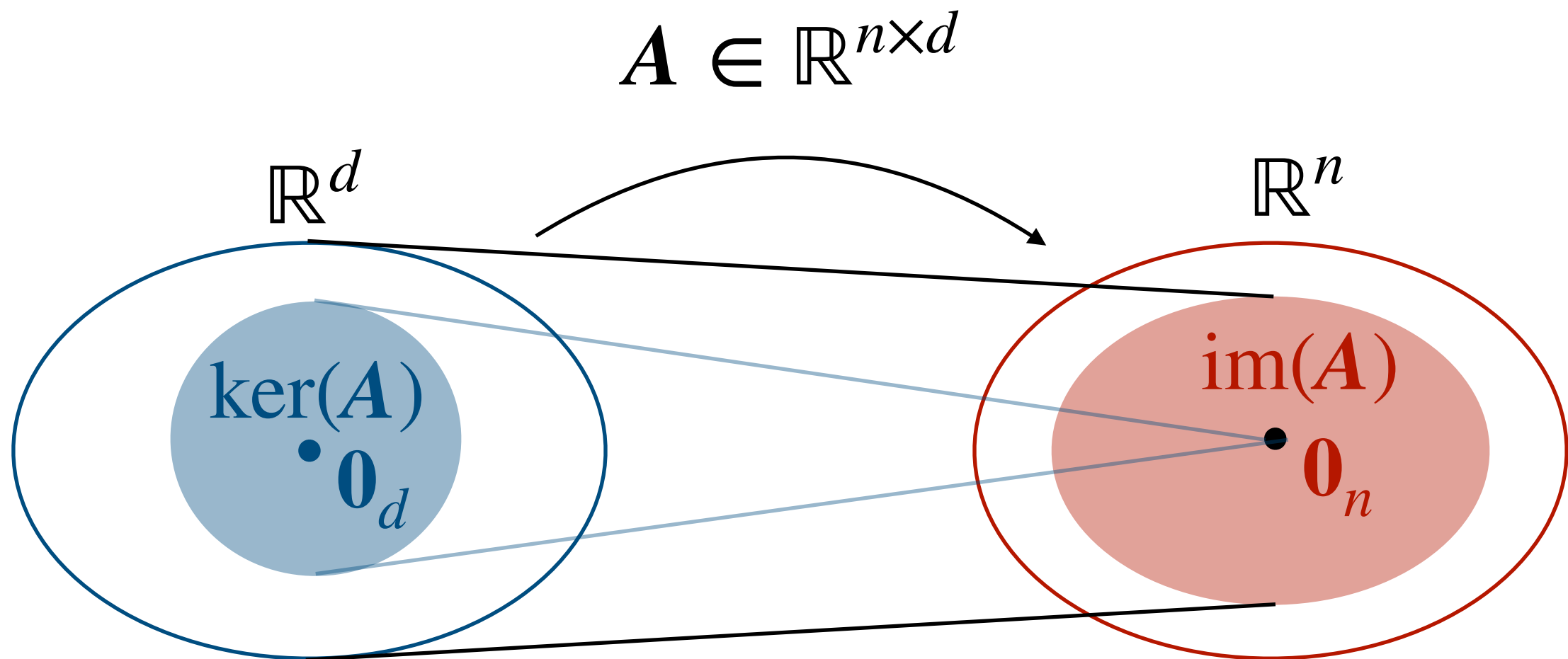


Note that  $\text{ker}(\mathbf{A}) \subset \mathbb{R}^d$   
and  $\mathbf{0} \in \text{ker}(\mathbf{A})$

# Image and null-space

## Proposition

Let  $A \in \mathbb{R}^{n \times d}$  denote a linear map. We have:  
$$\text{rank}(A) + \dim(\ker(A)) = d$$



# Matrix inverse

---

A square matrix  $\mathbf{A} \in \mathbb{R}^{d \times d}$  is said to be invertible if there exists  $\mathbf{B} \in \mathbb{R}^{d \times d}$  such that:

$$\mathbf{AB} = \mathbf{I}_d$$

In this case, we denote  $\mathbf{B} = \mathbf{A}^{-1}$ .

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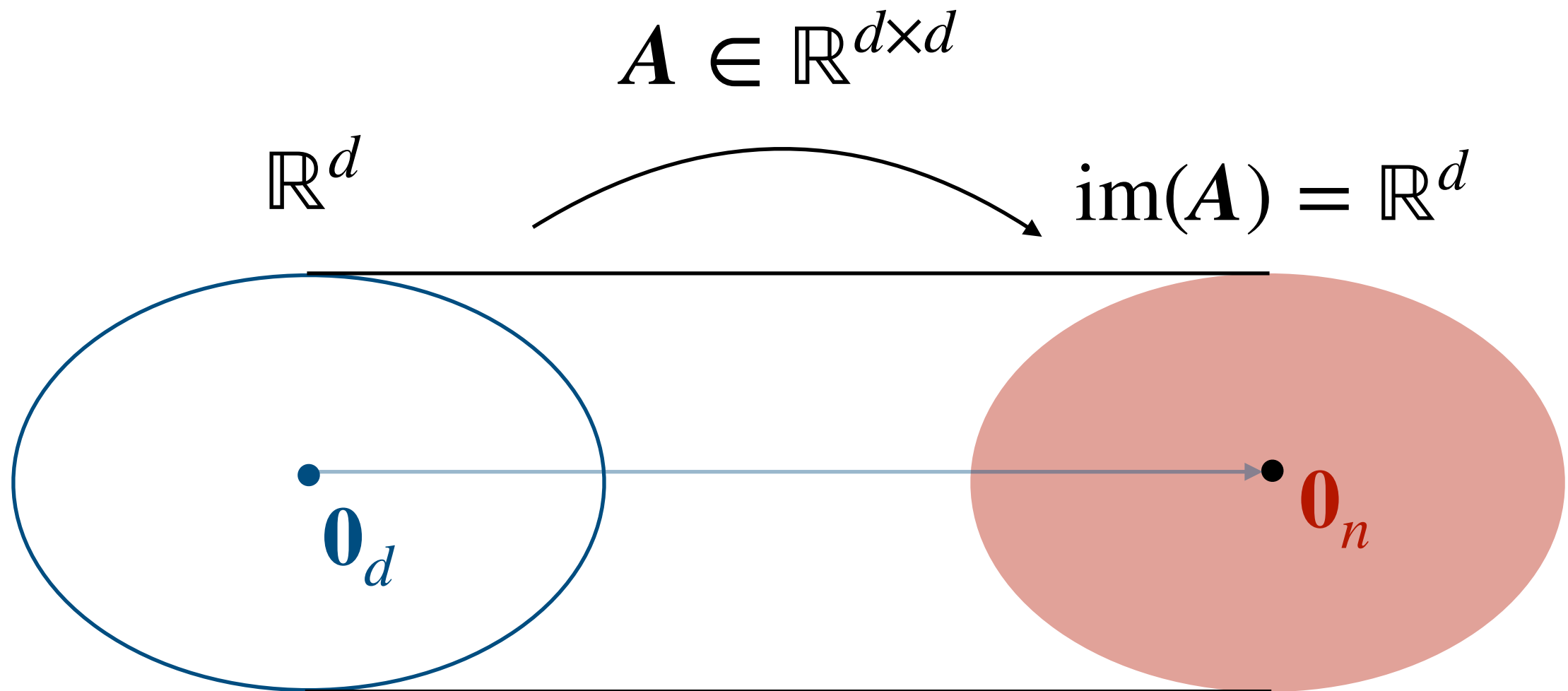
## Proposition

A square matrix  $\mathbf{A} \in \mathbb{R}^{d \times d}$  is invertible  
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# Matrix transpose

- The **transpose** of a matrix  $\mathbf{A} \in \mathbb{R}^{n \times d}$  with elements  $a_{ij}$  the matrix with  $\mathbf{A}^\top \in \mathbb{R}^{d \times n}$  with elements  $a_{ji}$

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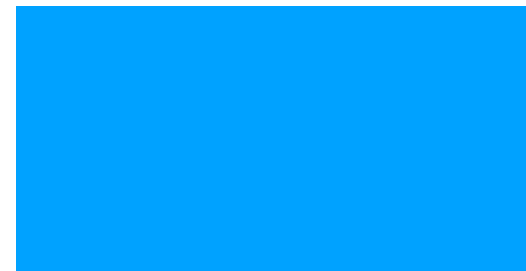
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- We have:

$$(\mathbf{A}^\top)^\top = \mathbf{A}$$

$$(a\mathbf{A} + b\mathbf{B})^\top = a\mathbf{A}^\top + b\mathbf{B}^\top$$

$$(\mathbf{A}^{-1})^\top = (\mathbf{A}^\top)^{-1}$$

$$(\mathbf{AB})^\top = \mathbf{B}^\top \mathbf{A}^\top$$



Exercise: check this.

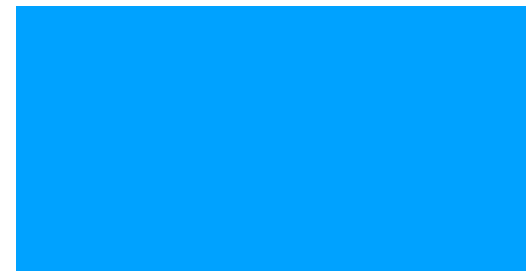
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- Note that by seeing  $\mathbf{u}, \mathbf{v} \in \mathbb{R}^{d \times 1}$  as column vectors, we can also write the Euclidean inner product as:

$$\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^\top \mathbf{v}$$



Exercise: check this.

# Matrix trace

---

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$$\text{Tr } \mathbf{A} = \sum_{i=1}^d a_{ii}$$

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- It satisfies:  $\text{Tr } \mathbf{AB} = \text{Tr } \mathbf{BA}$

$$\text{Tr } (a\mathbf{A} + b\mathbf{B}) = a\text{Tr } \mathbf{A} + b\text{Tr } \mathbf{B}$$

$$\text{Tr } \mathbf{A}^{\top} = \text{Tr } \mathbf{A}$$



Exercise: check this.

# Symmetric matrices

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Letting  $\mathbf{a}_i \in \mathbb{R}^d$  denote the rows of  $\mathbf{A} \in \mathbb{R}^{n \times d}$ , we have:

$$(\mathbf{A} \mathbf{A}^\top)_{ij} = \langle \mathbf{a}_i, \mathbf{a}_j \rangle$$



Exercise: check this.

Note: a similar representation holds for columns of  $\mathbf{A}$

# Orthogonal matrices

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$$||\mathbf{A}\mathbf{u}||_2 = ||\mathbf{u}||_2$$



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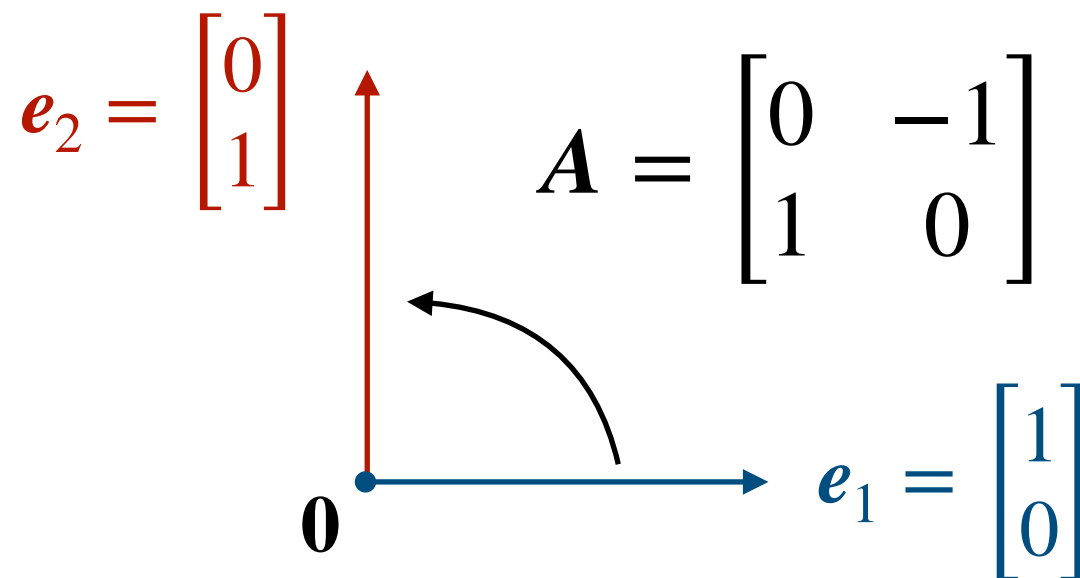
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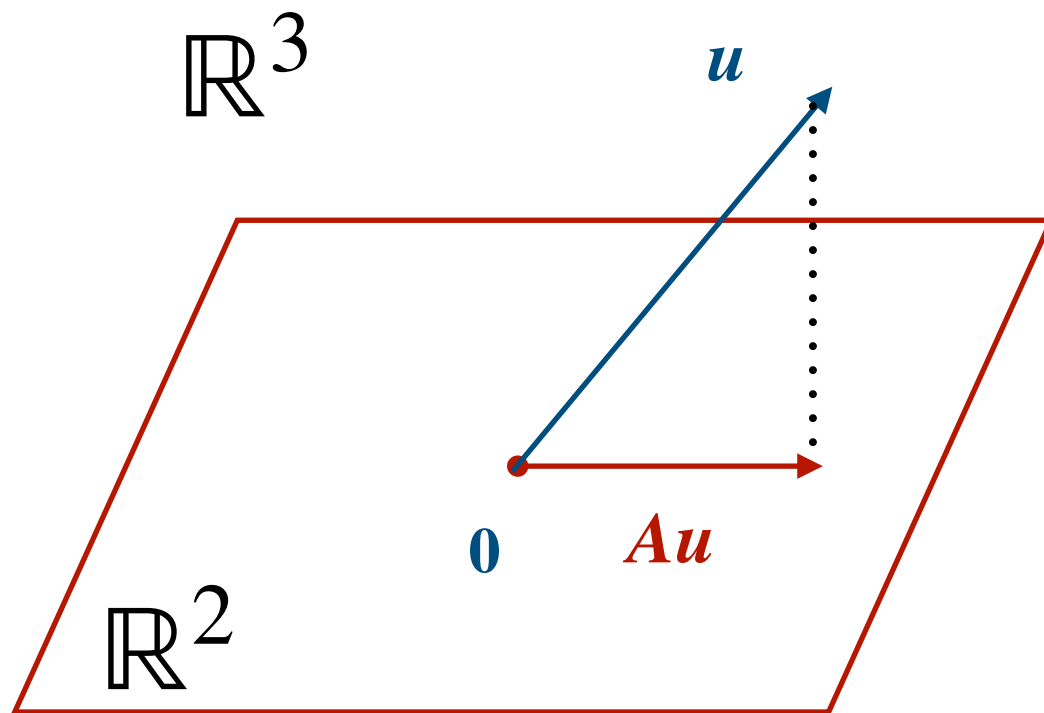
Geometrically, they define **rotations** and **reflections**



# Projection matrix

- A square matrix  $\mathbf{A} \in \mathbb{R}^{d \times d}$  is a **projection** if  $\mathbf{A}^2 = \mathbf{A}$

Moreover, if  $\mathbf{A}$  is also symmetric  $\mathbf{A} = \mathbf{A}^\top$ , we call it a **orthogonal projection**.



$$\mathbf{P} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

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## Proposition

Any  $\mathbf{v} \in \mathbb{R}^d$  can be uniquely written as:

$$\mathbf{v} = (\mathbf{v} - \mathbf{A}\mathbf{v}) + \mathbf{A}\mathbf{v} \quad \begin{array}{l} \mathbf{v} - \mathbf{A}\mathbf{v} \in \ker(\mathbf{A}) \\ \mathbf{A}\mathbf{v} \in \text{im}(\mathbf{A}) \end{array}$$

$$\mathbb{R}^d = \text{im}(\mathbf{A}) \oplus \ker(\mathbf{A})$$

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The only projection matrix which is invertible is the identity.

# Eigen-(values, vectors)

Let  $A \in \mathbb{R}^{d \times d}$  denote a square matrix. An **eigenvector** is a non-zero vector that is only rescaled under the action of  $A$ :

$$Av = \lambda v \quad v \neq 0$$

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- A square matrix  $\mathbf{A} \in \mathbb{R}^{d \times d}$  can have at most  $d$  independent eigenvectors.



- An eigenvalue  $\lambda$  can be associated to more than one independent eigenvector.

# Positive matrices

- A symmetric square matrix  $A \in \mathbb{R}^{d \times d}$  is called **positive-definite** if all eigenvalues are positive:

$$\lambda \in \text{spec}(A) \Rightarrow \lambda > 0$$

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For any rectangular matrix  $\mathbf{B} \in \mathbb{R}^{n \times d}$ ,  
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Exercise: prove this.

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not necessarily positive definite.



Exercise: prove this.

# Spectral theorem

## Theorem

Any symmetric matrix  $\mathbf{A} \in \mathbb{R}^{d \times d}$  can be decomposed as

$$\mathbf{A} = \mathbf{U} \mathbf{D} \mathbf{U}^\top$$

$\mathbf{U} \in \mathbb{R}^{d \times d}$  is an orthogonal matrices and  $\mathbf{D}$  is a diagonal matrix with elements given by the eigenvalues.

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$\mathbf{U} \in \mathbb{R}^{d \times d}$  are orthogonal matrices and  $\mathbf{D}$  is a diagonal matrix with elements given by the eigenvalues.

We can equivalently write the spectral decomposition as:

$$\mathbf{A} = \sum_{i=1}^{\text{rank}(\mathbf{A})} \lambda_i \mathbf{v}_i \mathbf{v}_i^\top$$

Where  $\mathbf{v}_i \in \mathbb{R}^d$  are orthonormal eigenvectors.

# Important facts

- The trace of a symmetric matrix  $\mathbf{A} \in \mathbb{R}^{d \times d}$  is the sum of its eigenvalues

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- The eigenvalues of a projection matrix  $\mathbf{P} \in \mathbb{R}^{d \times d}$  are 0 or 1

$$\mathbf{P} = \sum_{i=1}^{\text{rank}(\mathbf{P})} \mathbf{v}_i \mathbf{v}_i^\top$$



Exercise: show this.

Moreover,  $\mathbf{P} \in \mathbb{R}^{d \times d}$  is orthogonal if  $\mathbf{v}_i$  are orthogonal vectors.

# Singular value decomposition

Note that for any real matrix  $\mathbf{A} \in \mathbb{R}^{n \times d}$ ,  $\mathbf{A}^\top \mathbf{A} \in \mathbb{R}^{d \times d}$  and  $\mathbf{A} \mathbf{A}^\top \in \mathbb{R}^{n \times n}$  are symmetric matrices.

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Therefore,  $\mathbf{A}^\top \mathbf{A}$  and  $\mathbf{A}\mathbf{A}^\top$  can be diagonalised:

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Where:  $r = \text{rank}(\mathbf{A}^\top \mathbf{A}) = \text{rank}(\mathbf{A}\mathbf{A}^\top)$

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$$\lambda_i \geq 0$$

Therefore, defining the **singular values**  $\sigma_i = \sqrt{\lambda_i}$

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This can be equivalently written as:

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With:  $\mathbf{U} \in \mathbb{R}^{n \times n}$  and  $\mathbf{V} \in \mathbb{R}^{d \times d}$  orthogonal matrices

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Computationally, it is more efficient to define

$\mathbf{U} \in \mathbb{R}^{n \times r}$ ,  $\mathbf{V} \in \mathbb{R}^{d \times r}$  and  $\mathbf{D} \in \mathbb{R}^{r \times r}$

# Pseudo-inverse

The SVD allow us to define a generalised notion of matrix inverse. Let  $\mathbf{A} \in \mathbb{R}^{n \times d}$  with SVD:

$$\mathbf{A} = \sum_{i=1}^{\text{rank}(\mathbf{A})} \sigma_i \mathbf{u}_i \mathbf{v}_i^{\top}$$

The **pseudo-inverse**  $\mathbf{A}^+ \in \mathbb{R}^{d \times n}$  is defined via its SVD:

$$\mathbf{A}^+ = \sum_{i=1}^{\text{rank}(\mathbf{A})} \frac{1}{\sigma_i} \mathbf{v}_i \mathbf{u}_i^{\top}$$

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It satisfies:  $\mathbf{A}\mathbf{A}^+\mathbf{A} = \mathbf{A}$      $\mathbf{A}^+\mathbf{A}\mathbf{A}^+ = \mathbf{A}^+$

$$(\mathbf{A}^+)^+ = \mathbf{A}$$

If  $\mathbf{A}$  is invertible,  $\mathbf{A}^+ = \mathbf{A}^{-1}$

If  $\mathbf{A}$  is full-rank, 
$$\mathbf{A}^+ = \begin{cases} (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top & \text{if } n \geq d \\ \mathbf{A}^\top (\mathbf{A} \mathbf{A}^\top)^{-1} & \text{if } n < d \end{cases}$$



Exercise: show this.

# Pseudo-inverse

The pseudo-inverse is useful to define orthogonal projectors

For any real matrix  $\mathbf{A} \in \mathbb{R}^{n \times d}$ :

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Similarly,

$$\mathbf{I}_d - \mathbf{A}^+ \mathbf{A} \in \mathbb{R}^{d \times d}$$

$$\mathbf{I}_n - \mathbf{A} \mathbf{A}^+ \in \mathbb{R}^{n \times n}$$

Define orthogonal projection operators in the kernel of  $\mathbf{A}$  and  $\mathbf{A}^T$ , respectively.

# Recap of Probability

The butter of statistical learning

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- **Discrete**: when the possible outcomes are **countable**.

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- rolling a die  $X \in \{1, \dots, 6\}$
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Discrete r.v.s are described by their probability distribution

$$\mathbb{P}(X = k)$$

A non-negative “function” that sums to one.

$$\sum_{k \in \text{supp}(X)} \mathbb{P}(X = k) = 1$$

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Continuous r.v.s are described by their probability density function (p.d.f.), which integrates to probabilities:

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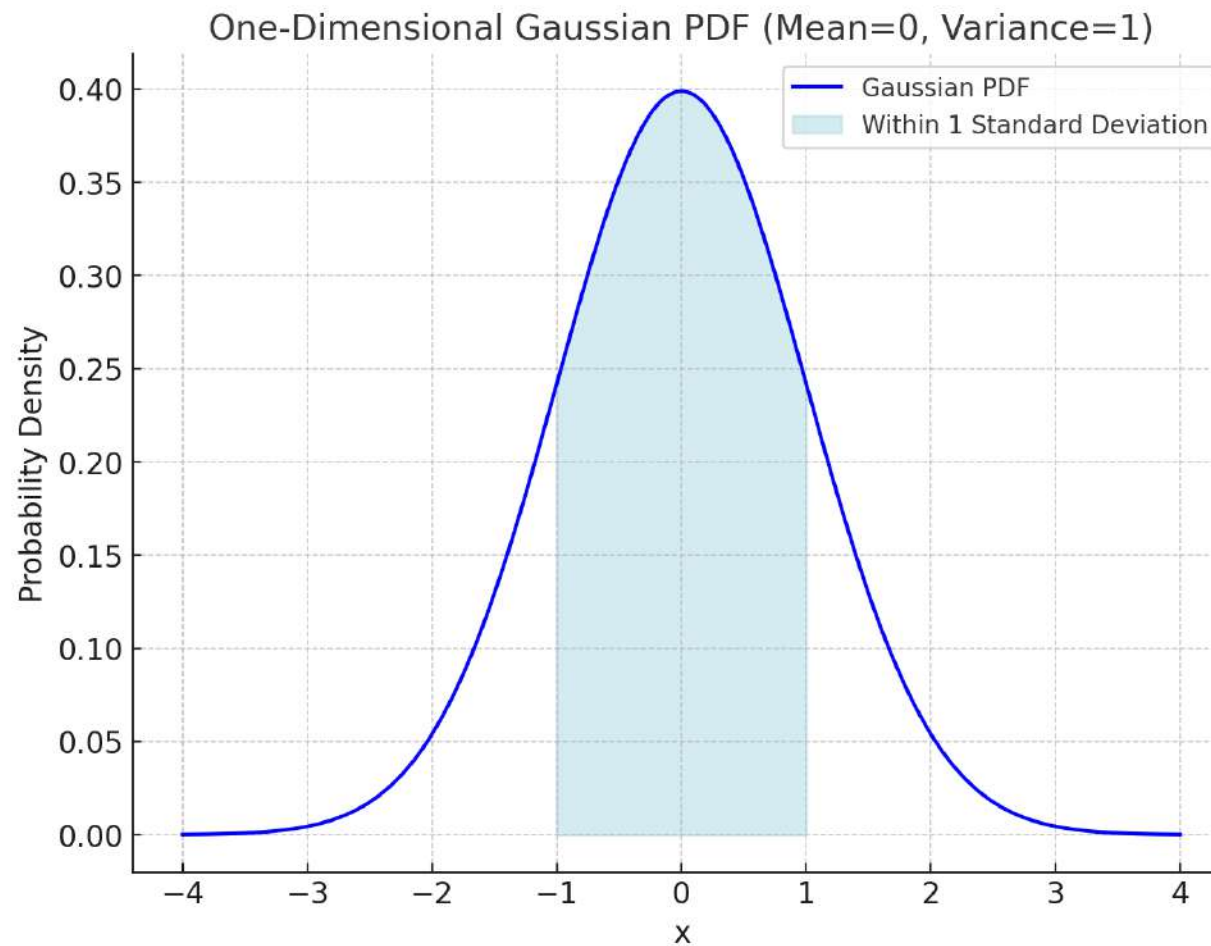


The p.d.f. is **NOT a probability**. It can be  $p_X(x) \geq 1$ .

# Normal distribution

A Gaussian r.v.  $X \sim \mathcal{N}(\mu, \sigma^2)$  has the following p.d.f.:

$$p_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

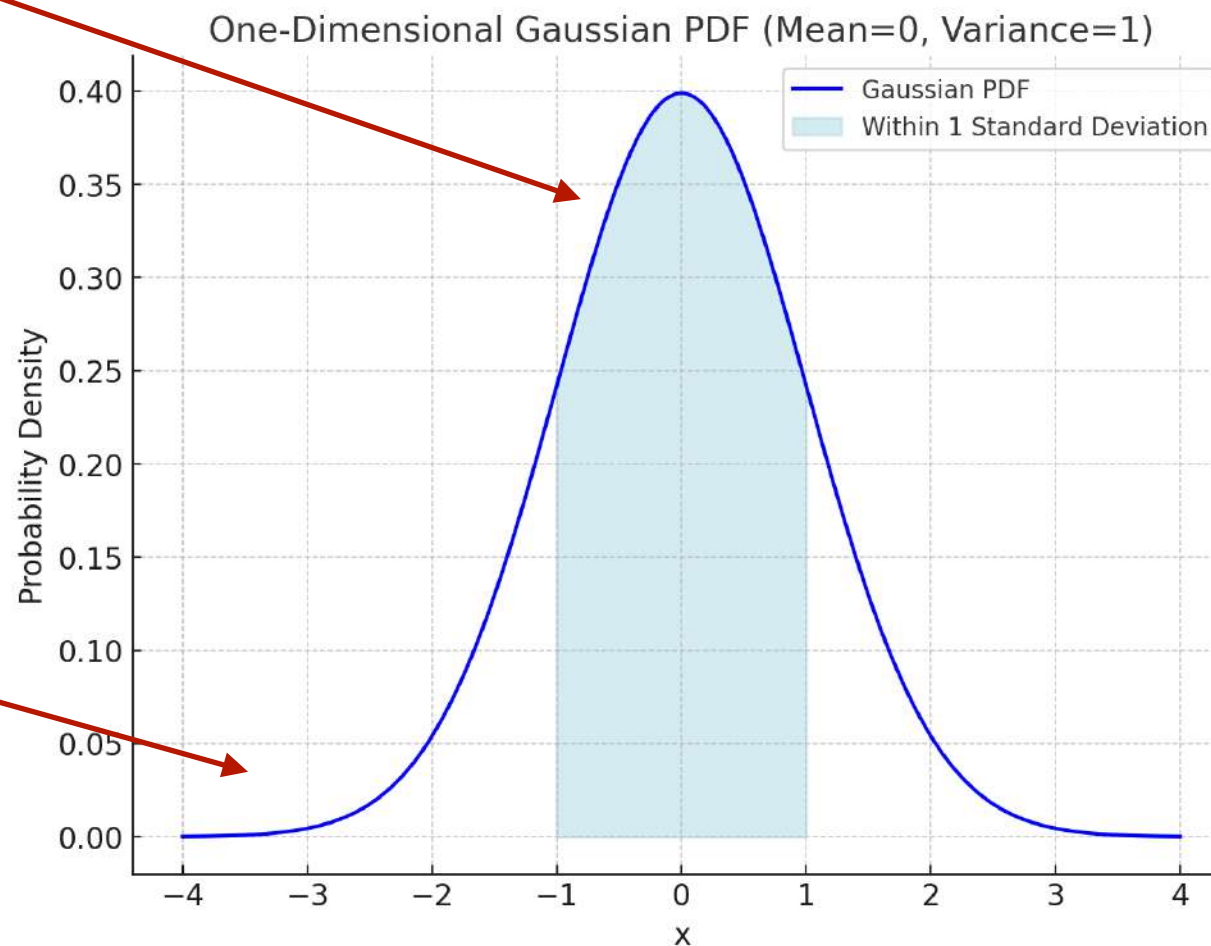


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High-probability



Low-probability

# Expectation and variance

Let  $X \sim p_X$  denote a continuous r.v.

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- The **variance** of  $X$  is defined as:

$$\text{Var}[X] = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$

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# Change of variables

Let  $X \sim p_X$  denote a continuous r.v. and  $f: \mathbb{R} \rightarrow \mathbb{R}$

Then,  $Y = f(X)$  is also a random variable, with p.d.f. given by

$$p_Y(y) = \int dx \, p_X(x) \delta(y - f(x))$$

Where  $\delta(x)$  is the “Dirac delta function”:

$$\int_{\mathbb{R}} dx \, \delta(x - y) f(x) = f(y)$$

# Joint distribution

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We say  $X, Y$  are **uncorrelated** if  $\text{Cov}(X, Y) = 0$   
( $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$ )

# Independence

- Given two r.v.s  $X, Y \sim p_{X,Y}$ , we define the **marginal distributions**

$$p_X(x) = \int dy \, p_{X,Y}(x, y) \qquad p_Y(y) = \int dx \, p_{X,Y}(x, y)$$

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Note that independence implies uncorrelated, but not the converse!



Exercise: Construct a counterexample

# Conditional distribution

- Given two r.v.s  $X, Y \sim p_{X,Y}$ , we define the conditional p.d.f.

$$p_{X|Y}(x|y) = \frac{p_{X,Y}(x,y)}{p_Y(y)} \quad (p_Y(y) > 0)$$

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Theorem (Bayes theorem)

$$p_{X|Y}(x|y) = \frac{p_{Y|X}(y|x)p_X(x)}{p_Y(y)}$$

# Law of large numbers

Let  $X_1, \dots, X_n \sim p_X$  denote i.i.d. r.v.s. with mean  $\mathbb{E}[X_i] = \mu$

Define the sample mean (note this is itself a r.v.)

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

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Theorem (Weak LLN)

$$\bar{X}_n \xrightarrow{P} \mu \quad \text{as } n \rightarrow \infty$$

$$\forall \epsilon > 0 \quad \lim_{n \rightarrow \infty} \mathbb{P}(|\bar{X}_n - \mu| < \epsilon) = 1$$



Be aware there are many variations of the LLN.

# Central limit theorem

---

Let  $X_1, \dots, X_n \sim p_X$  denote i.i.d. r.v.s. with mean  $\mathbb{E}[X_i] = \mu$  and variance  $\text{Var}(X_i) = \sigma^2 < \infty$

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Theorem (Lindeberg–Lévy CLT)

$$\sqrt{n} \frac{\bar{X}_n - \mu}{\sigma} \xrightarrow{d} \mathcal{N}(0,1)$$

$$\lim_{n \rightarrow \infty} \mathbb{P}(\sqrt{n}(\bar{X}_n - \mu) \leq z) = \mathbb{P}(Z \leq z/\sigma) \quad Z \sim \mathcal{N}(0,1)$$



Be aware there are many variations of the CLT.

99% of books in statistical learning:

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